United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Mid-Term Exam: Trimester: Summer 2024

Course Code: EEE 2123; **Course Title:** Electronics
Total Marks: 30; **Duration:** 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Q1) For the following graph :

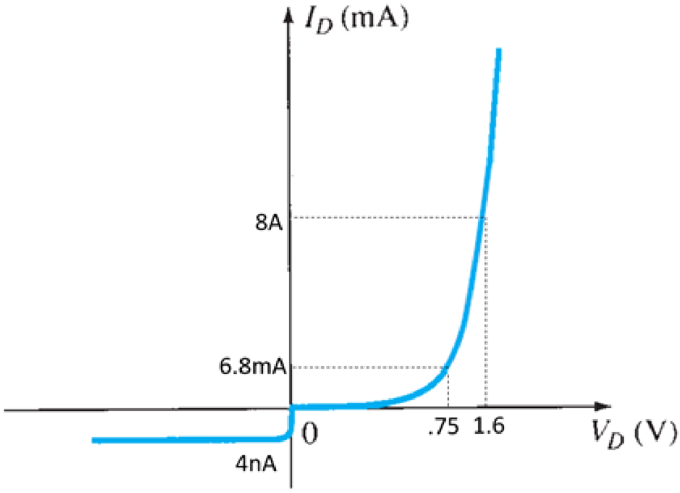

Fig 1: Diode I/V graph for Q1

a. The reverse saturation current is shown at 18°*C*. Find the I_s when temperature increased to 85°C. [3]

b. Find the static resistance when the biasing voltage is at -30v at 0°C. [1]

c. Draw the approximate I/V curve for the operating temperature of 85°C with reference to the given I/V curve. [2]

Q2. **a)** Find I_1, I_2, I_3, V_1, V_2 & V_0 in the following circuit. [5]

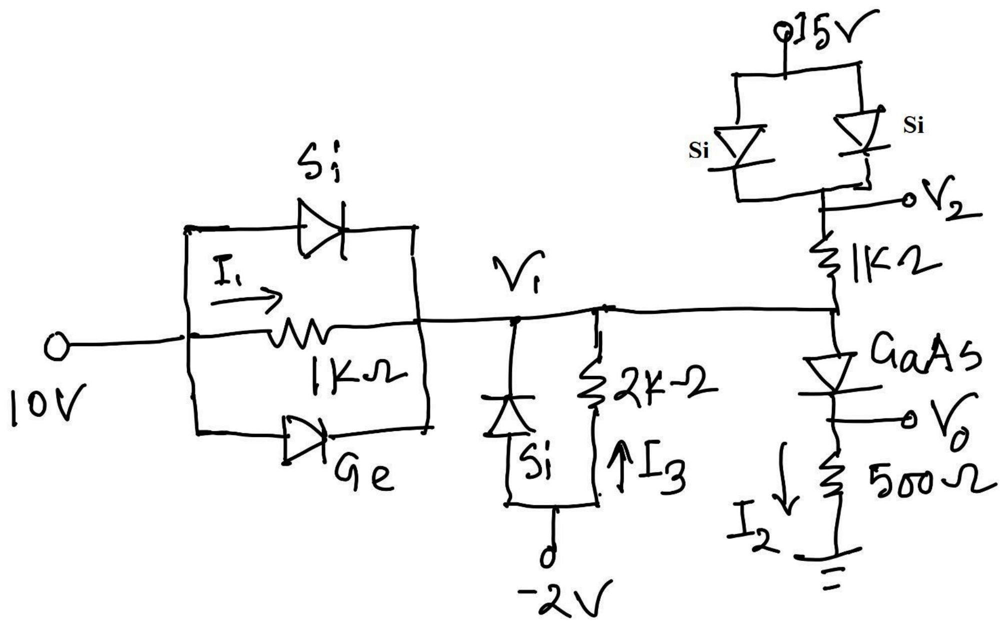

Fig 2: Circuit diagram for Q2(a)

b) Will both the diodes be on in the following circuit? Also, find the value of V and I. [4]

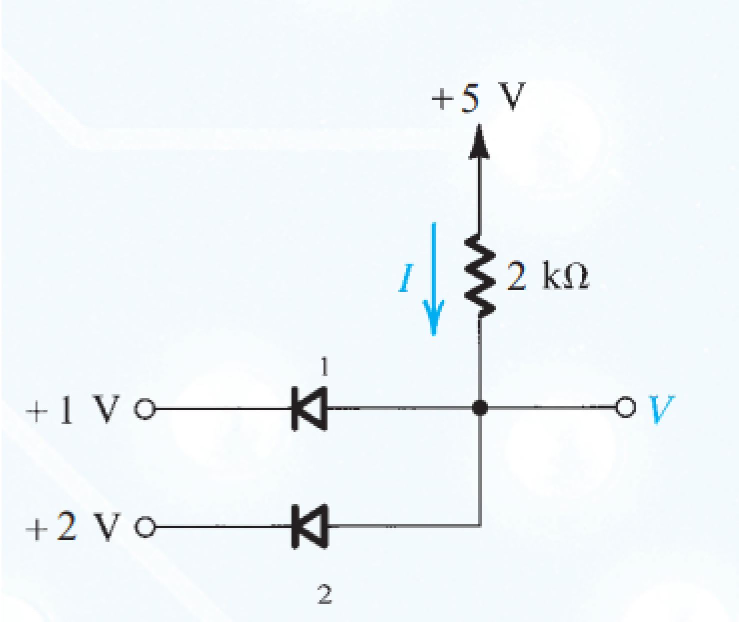

Fig 3: Circuit diagram for Q2(b)

Q3. Consider the following circuit in Fig 4. The input is a sine wave given by $\boldsymbol{v_i = 10 \sin (2\pi t)}$. Assume diodes are ideal.

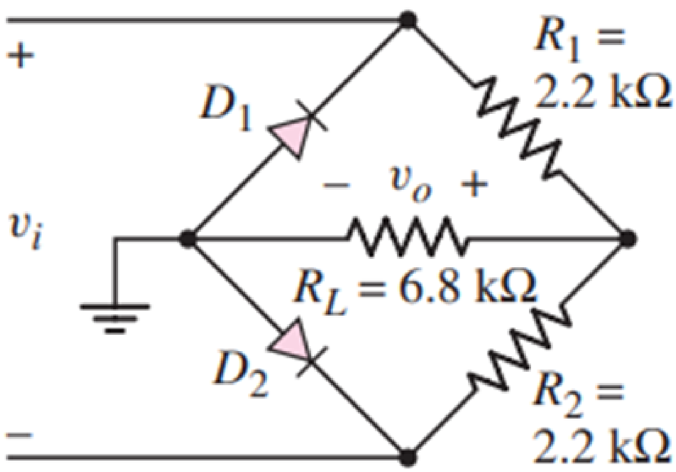

Fig 4: Circuit diagram for Q3

a) Derive the output voltage expression and sketch proper labels and peak values in your graph. [3.5]

b) If the diodes can withstand a maximum reverse voltage of 15 V, then explain whether the circuit is safe or not. [1.5]

c) Find average values of both input and output voltages. [1]

Q4. Analyze and sketch V_o for the network of the following circuit for the input shown below. Clearly mention the peak values in your sketch. [4]

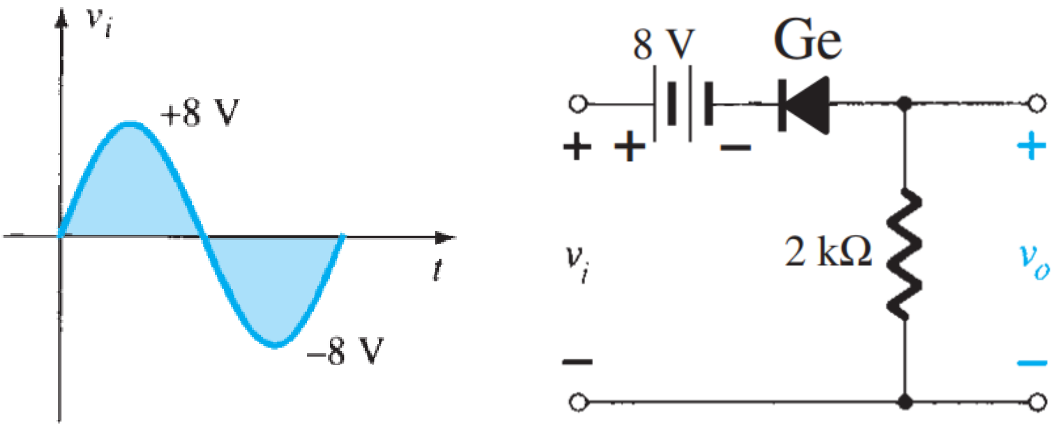

Fig 5: Circuit diagram for Q4

Q5. Design the clipper and clamper circuit to produce the following output voltage (v_{o2}) according to the given input voltage (v_i). In the following network, the input and output of the clamper circuit are v_i and v_{o1} . For the clipper circuit, the input and output are v_{o1} and v_{o2}. For the clamper circuit, use Ge diodes/diode and for the clipper circuit, use GaAs diode/diodes. [5]

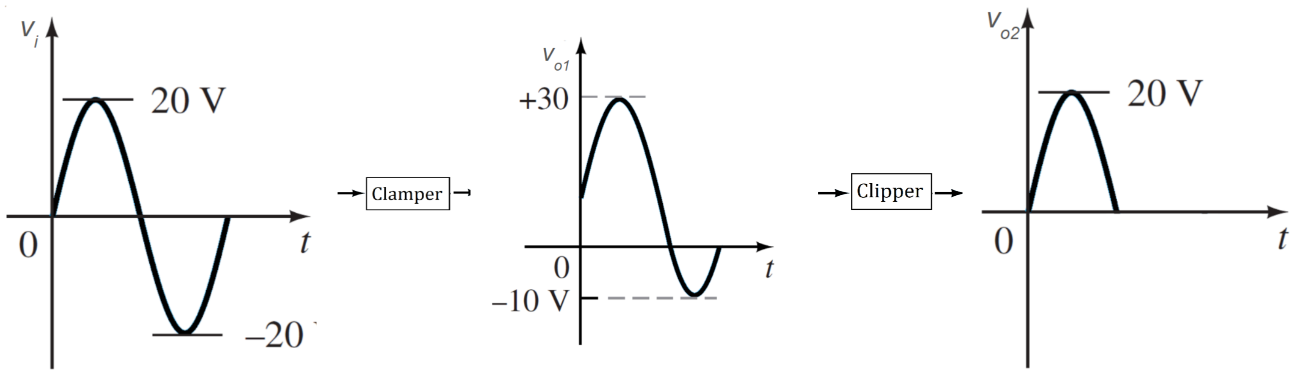

Fig 6: Diagram for Q5